

Course Syllabus

STAT 109: Introductory Biostatistics

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Cal Poly Humboldt — Spring 2026

Course Information

Course Number: STAT 109

Course Title: Introductory Biostatistics

Units: 4

Semester: Spring 2026

Mode of Instruction: In-person

General Education: Lower Division GE Area B4 – Mathematical Concepts & Quantitative Reasoning

Lecture

Days/Time: Monday, Wednesday, Friday, 11:00–11:50 AM

Location: BSS 166

Laboratory (Thursday: enroll in one section)

9:00–10:50 AM, LOCATION: FH 202

11:00–12:50 PM, LOCATION: FH 202

1:00–2:50 PM, LOCATION: FH 202

Final Exam (Optional)

Optional Final Exam (held during the scheduled final exam period): Wednesday, May 13, 2026, 10:20–12:10 PM, BSS 166. The final exam is optional and may be taken to replace a midterm grade.

Instructor Information

Teacher: Dr. Rosanna Overholser

Email: rho3@humboldt.edu

Office Location: BSS 334

Office Phone: 707-826-4022

Office Hours: Mondays 2-3pm, Wednesdays 12-12:50pm, and Fridays 10-10:50am or by appointment

Please use your Cal Poly Humboldt email account to contact me and check both Canvas and your Cal Poly Humboldt email regularly for course communications.

Course Description (Catalog Description)

Descriptive statistics, probability, random variables, discrete and continuous distributions, confidence intervals, contingency tests, regression and correlation, tests of hypothesis, analysis of variance. Emphasis: methods and applications used in the biological and natural resource sciences.

Division: Lower Division **Weekly:** Lecture 3 hrs. Activity 2 hrs. **Instruction Mode:** In-person

Prerequisite: MATH 101 or MATH 101I or MATH 102 or MATH 109B (may be taken concurrently with instructor

approval) with a grade of C- or better or ALEKS PPL assessment score over 65 or appropriate high school coursework. See [Math Placement and ALEKS PPL](#) for details. MATH 102 may be taken concurrently with instructor approval.

Note: if you plan to use STAT 109 as a prerequisite for another MATH, DATA or STAT course, you must have a grade of C- or better in this course.

Course Learning Outcomes

By the end of this course, students will be able to:

1. Use numerical and graphical descriptive statistics to summarize and interpret data.
2. Explain the role of randomness, probability, and sampling in statistical inference.
3. Apply probability rules and work with common probability distributions.
4. Construct and interpret confidence intervals for means and proportions.
5. Conduct and interpret basic hypothesis tests.
6. Analyze relationships between pairs of variables using correlation and regression.
7. Use a statistical programming language to analyze data and communicate results.
8. Distinguish between correlation and causation and recognize limitations of statistical conclusions.

General Education & Institutional Learning Outcomes

This course fulfills **Lower Division General Education Area 2 (Quantitative Reasoning)** and supports Cal Poly Humboldt's Institutional Learning Outcomes related to quantitative reasoning, critical thinking, and communication.

Course Materials & Workload

STAT 109 includes:

- Three weekly lecture meetings
- One weekly laboratory session

In this 4-unit course, students should expect to spend approximately **12 hours per week** on course-related work, including time in lecture (3 hours), lab time (2 hours), and time outside of class to do practice problems, projects and prepare for exams and quizzes (7 hours).

Required Materials

Lecture Notes: I'll provide packets of guided notes to you in class. These notes have key concepts, definitions and word problems that we'll solve during class time. These notes have spaces for you to fill in during lecture as we go. If you need to miss class, please ask me for the packet you will be missing.

Optional Textbook: [Introductory Statistics for the Life and Biomedical Sciences](#) by Julie Vu and Dave Harrington, available for free online.

Three Binder: Please bring a three-ring binder to class to keep the lecture note packets in.

Pen or Pencil: Please bring a pen or pencil to class to take notes with.

Software: R (free and installed on all campus computers; see [R download page](#) to install on your own computer)

Calculator: Scientific (not graphing) calculator

Other Technology: Regular access to a computer with reliable internet

Access & Affordability

Students who do not have personal access to required technology may use campus computer labs and library resources. If you want to get a free used computer, please check to see if our campus's [Reuseable Office Supply Exchange \(ROSE\)](#) has one available.

Fees

There are no additional course fees beyond standard tuition and registration fees.

Course Feedback & Assessment

Students will complete a variety of assessments:

- Daily warm-up problems (Mondays, Wednesdays and Fridays, in lecture)
- Weekly programming labs (due Tuesdays, in Canvas: I'll help you write the code the Thursday before)
- Weekly quizzes based on practice problems and labs (Thursdays at the beginning of lab)
- Two midterms (Week 6 and Week 13 Fridays, in lecture)
- Five projects (due on assigned dates in Canvas; see important dates below)

Daily warm-up problems are designed to give you an opportunity to work as a team with your classmates. Weekly labs and quizzes are designed to provide regular formative feedback through practice problems, posted solutions, and in-lab support. Midterms require students to demonstrate their understanding of the course material. Projects allow students to apply their knowledge to practical, real-world problems and can be used as part of a portfolio.

Optionally, students may take a final exam (held during the scheduled final exam period: Wednesday, May 13, 2026, 10:20–12:10 PM, BSS 166) to replace a midterm grade.

Grading

Component	Weight
Daily Participation	5%
Weekly quizzes	30%
Weekly labs	15%
Two midterms	30%
Five projects	20%
Total	100%

Final letter grades will be assigned based on overall performance according to the following:

A: 93-100% A-: 90-92% B+: 87-89% B: 83-86% B-: 80-82% C+: 77-79% C: 73-76% C-: 70-72% D: 60-69% F: 0-59%

Note:

- **Grade Mode:** Letter grade by default. Students may elect Credit/No Credit grading according to university deadlines and procedures.
- A minimum grade of **C-** is required for the course to count toward GE Area B4.
- If you stop attending class and stop doing assignments without communicating with me, you will receive a grade of WU (Withdrawal Unsatisfactory). Please communicate with me early if you are having difficulties: it's possible that I can help you get back on track, or help you obtain a more appropriate grade (W or I) for your situation.

Attendance & Participation

Regular attendance and active participation are expected and essential for success in this course.

- Attendance in lecture will be tracked via participation in the warm-up problems.
- Attendance in lab will be tracked via participation in the quizzes.

If you must miss class, you are responsible for obtaining notes and keeping up with course material.

Late or Missed Work Policies

The policies for late or missed work are as follows:

- **Daily attendance** in lecture is required, but I know that sometimes you will need to miss class due to illness or other emergencies. Each student may miss up to 10 days of lecture without penalty and without contacting me. It's up to you to reserve these days for emergencies only. In my experience, it is rare for a student to succeed if they regularly miss lecture or are absent for more than two weeks so if you find yourself in this situation, please communicate with me and your advisor so we can help you find a solution to getting back on track.
- **Weekly quizzes** during Thursday lab sessions are required, but I know that sometimes you will need to miss lab due to illness or other emergencies. If this happens, please contact me to make an arrangement to take the quiz at another time.
- **Weekly programming labs** are due Tuesdays in Canvas. After the due date, a solution will be posted, so late submissions are generally not accepted except in documented emergencies. These labs will be graded on completion and your reflection.
- **Projects** are due on the specified date in Canvas. You may submit a late project up to a week late with no penalty. After a week, a project can be turned in for 70% of the grade.
- **Midterms** are required. If you need to miss a midterm due to illness or other emergencies, please communicate with me as soon as possible to reschedule it. Alternatively, you may take the optional final exam (Wednesday, May 13, 2026, 10:20–12:10 PM, BSS 166) to replace a missed midterm grade.

Note: Technology issues (internet access, device failure, software problems) are not automatically considered valid excuses for late or missed work; students are expected to plan accordingly.

AI Policy

Much of your course grade will be based on materials that you produce in-class without having AI as a resource. But you may use AI, such as NotebookLM or ChatGPT, in this course to help you learn the materials or to perform data analyses or on the projects.

For work that you do at home that is turned in for a grade I have two requirements:

- 1) if you used AI, acknowledge this briefly in the document (you don't need to state in detail how you used AI or give your prompts, just the name of the AI service you used).
- 2) Be responsible for the correctness and appropriateness of all (AI assisted or not) materials you turn in.

I may use AI to assist me in my delivery of the course material, such as asking AI to see if an assignment is appropriate for a college level course or if the wording of a quiz question could be improved. I will not use AI to grade your work or use an AI detection tool to check your work.

Course Topics & Schedule

Topics (Approximate Order)

- Introduction to data, simulation, and statistical reasoning
- Selected topics in probability
- Binomial Random Process
- A simple hypothesis test
- Normal probability distribution
- Sampling distributions and the Central Limit Theorem

- Sampling methods
- Confidence intervals
- Study design
- More Hypothesis testing
- Simple predictive modeling via Linear regression

Summary of Schedule and Important Dates

Note: This schedule is subject to change. Any changes will be communicated through **Canvas**.

Date	Event
January 20	Holiday (No class)
January 23	Quiz 1 (during lab)
February 2	Add/Drop Deadline without a serious and compelling reason
February 22	Project 1 due
February 27	Midterm 1
March 15	Project 2 due
March 16-20	Spring Break (No classes)
March 31	Cesar Chavez Day (No class)
April 5	Project 3 due
May 8	Project 4 due
April 24	Midterm 2
May 15	Project 5 due
May 13	Final Exam (Optional, to replace a midterm grade) - 10:20 AM-12:10 PM, BSS 166

Weekly Schedule:

- **Quizzes:** Weekly quizzes (Quiz 1-12) on Thursdays during lab sessions
- **Lectures:** Monday, Wednesday, Friday, 11:00-11:50 AM with warm-ups in the first 5 minutes of class
- **Labs:** Get help on the code for your programming lab on Thursday (select one section: 9:00-10:50 AM, 11:00-12:50 PM, or 1:00-2:50 PM), then turn into Canvas by the following Tuesday. Elements of a lab may appear on the quiz, in addition to questions from that week's practice problem set.

Supplemental Instruction (SI)

This class is accompanied by a 1-unit Supplemental Instruction course designed to assist students in better understanding course materials. There is room for 20 students within the SI course, and it can be accessed through the normal enrollment process.

- **SI Course name:** Stat 198 - 31 Supplemental Instruction
- **Class number:** 22835
- **Time and place:** TR 4:00 - 4:50 PM FH 179
- **Instructor:** Em Madrid
- **Contact:** ebm37@humboldt.edu

University Policies & Resources

Students are responsible for reviewing the [Cal Poly Humboldt Syllabus Addendum](#), which includes policies and resources related to:

- Academic honesty
- Disability accommodations

- Emergency procedures
- Title IX
- Student conduct and support services

Accommodations

Students who wish to request disability-related accommodations should contact the Student Disability Resource Center (SDRC) as early as possible and then email me (Rosanna Overholser, rho3@humboldt.edu) your letter of accommodation.

Classroom & Online Conduct

Standards of respectful and professional behavior apply to all course interactions, including in-person classes, assignments, and email communication.

Changes to the Syllabus

This syllabus is subject to change. Any changes will be communicated through **Canvas announcements**.
